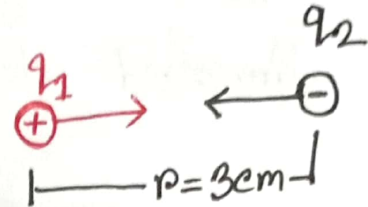


ELECTRIC FIELDS

Force between two-point charges

1. Given,



$$q_1 = 25 \text{ nC}$$

$$= 25 \times 10^{-9} \text{ C}$$

$$q_2 = -75 \text{ nC}$$

$$= -75 \times 10^{-9} \text{ C}$$

$$r = 3 \text{ cm} = 0.03 \text{ m}$$

Force on q_1 due to q_2

$$(b) \quad F_{12} = \frac{k q_1 q_2}{r^2}$$

$$= \frac{(9 \times 10^9) \times (25 \times 10^{-9}) \times (75 \times 10^{-9})}{(0.03)^2}$$

$$F_{12} = 0.01875 \text{ N} \leftarrow \begin{array}{l} \text{magnitude of force} \\ \text{exerted by } q_2 \text{ on } q_1 \end{array}$$

Since both the charges are opposite in sign they attract each other. So, F_{12} is directed towards q_2 .



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P.T.O

(a) As, there are only 2 charges involved, we get

$F_{21} = 0.01875 \text{ N}$ ← magnitude of force exerted by q_1 on q_2 and will be directed towards q_1 .

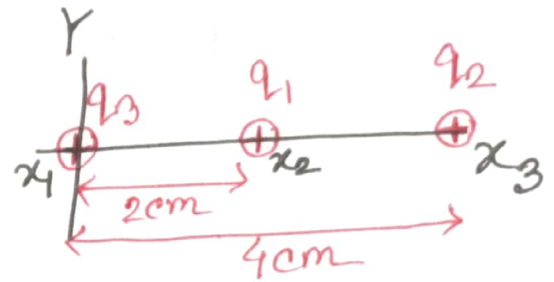
2. Given,

$$q_1 = 1 \text{ nC} = 1 \times 10^{-9} \text{ C}$$

$$q_2 = -3 \text{ nC} = -3 \times 10^{-9} \text{ C}$$

$$q_3 = 5 \text{ nC} = 5 \times 10^{-9} \text{ C}$$

$$x_1 = 0 ; x_2 = 2 \text{ cm} ; x_3 = 4 \text{ cm} \\ = 0.02 \text{ m} \quad \quad = 0.04 \text{ m}$$



The total electric force on the third charge q_3 caused by charges q_1 and q_2 is the sum of the forces of both charges (1 and 2) acting on the charge

$$3. \quad F_T = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_3}{(x_2 - x_1)^2} + \frac{1}{4\pi\epsilon_0} \frac{q_2 q_3}{(x_3 - x_1)^2}$$

$$F_T = 1.12 \times 10^{-4} \text{ N}$$

Electric field

Date / /

1.

$$E_C = E_1 + E_2$$

$$= \frac{-kq_1}{Ac^2} + \frac{kq_2}{Bc^2}$$

$$= k \left[\frac{q_2}{Bc^2} - \frac{q_1}{Ac^2} \right]$$

$$= 9 \times 10^9 \left[\frac{15 \times 10^{-9}}{(0.005)^2} \right.$$

$$\left. - \frac{30 \times 10^{-9}}{(0.02)^2} \right]$$

$$= 4.725 \times 10^6 \frac{V}{m}$$

$$q_1 = 30 nC$$

$$q_1 = 30 nC$$

$$= 30 \times 10^{-9} C$$

$$q_2 = 15 nC$$

$$= 15 \times 10^{-9} C$$

$$Bc = 5 mm$$

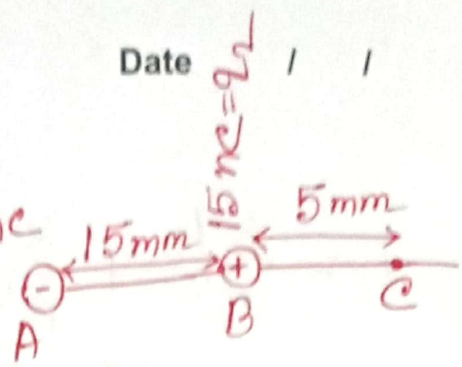
$$= 0.005 m$$

$$AB = 15 mm$$

$$= 0.015 m$$

$$Ac = 20 mm$$

$$= 0.02 m$$



2. $E_C = E_1 + E_2$

$$= k \left[\frac{q_1}{Ac^2} + \frac{q_2}{Bc^2} \right]$$

$$= 9 \times 10^9 \left[\frac{30 \times 10^{-9}}{(0.15)^2} + \frac{30 \times 10^{-9}}{(0.05)^2} \right]$$

$$= 1.2 \times 10^5 \frac{V}{m}$$

$$A \oplus \xleftarrow{10 cm} B \oplus \xrightarrow{5 cm} C$$

$$q_1 = 30 nC \quad q_2 = 30 nC$$

$$Ac = 15 cm = 0.15 m$$

$$Bc = 5 cm = 0.05 m$$



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3.

$$E = \frac{kq_1}{r_1^2} + \frac{kq_2}{r_2^2}$$

$$= k \left[\frac{q_1}{r_1^2} - \frac{q_2}{r_2^2} \right]$$

$$= 9 \times 10^9 \left[\frac{10 \times 10^{-9}}{(0.5)^2} - \frac{5 \times 10^{-9}}{(0.5)^2} \right]$$

$$= 180 \frac{V}{m}$$

$$q_1 = 10nC$$

$$q_2 = -5nC$$

